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EE1060: Differential Equations and Transform Techniques

Lecture 14: Solution of Second-order Homogeneous ODEs

Topics :

1. General solution of Second-order Homogeneous ODE
2. Solution of Second-order Homogeneous ODE with constant coefficients

General solution of Second-order Homogeneous ODE

$y'' + P(x)y' + Q(x)y = 0 \rightarrow (1)$ and $y_1(x)$ is known to be a solution.

goal: Construct the general solution $y(x) = c_1 y_1(x) + c_2 y_2(x)$

$\hookrightarrow$ Need to compute $\downarrow$ must be LI of $y_1(x)$

Key idea:- Construct $y_2(x) = v(x) y_1(x) \rightarrow (2)$ if $v(x) \neq c$, then y_2 & y_1 are LI.

find $v(x)$ such that $y_2(x)$ is a solution.

$y_2' = v' y_1 + v y_1' \rightarrow (3)$ and $y_2'' = v'' y_1 + 2v' y_1' + v y_1'' \rightarrow (4)$

Sub (3) & (4) in (1) $\Rightarrow$ $v'' y_1 + 2v' y_1' + v y_1'' + P(x)[v' y_1 + v y_1'] + Q(x) v y_1 = 0.$

$$\underbrace{v y_1'' + v P(x) y_1' + v Q(x) y_1}_{v\left(y_1'' + P(x) y_1' + Q(x) y_1\right) \,=\, 0} + v'' y_1 + v'(2y_1' + P(x) y_1) = 0$$

$$v'' + v'\left(2\frac{y_1'}{y_1} + P(x)\right) = 0 \Rightarrow \frac{v''}{v'} = -2\frac{y_1'}{y_1} - P(x) \rightarrow (5)$$

Integrating (5) on both sides

$$\ln(v') = -2\ln(y_1) - \int P(x)\,dx.$$

General solution of Second-order Homogeneous ODE

$$v' = \frac{1}{y_1^2} e^{-\int P(x)\,dx} \longrightarrow (6)$$

Integrating (6) on both sides

$$v = \int \underbrace{\left(\frac{1}{y_1^2} e^{-\int P(x)\,dx}\right)} dx \longrightarrow (7)$$

Example

Compute the general solution of $(1-x^2)y'' - 2xy' + p(p+1)y = 0$ with $p=1$. Given $y_1 = x$. (Legendre's Equation).

Normal form $y'' - \underbrace{\frac{2x}{1-x^2}}_{P(x)} y' + \underbrace{\frac{2}{(1-x^2)}}_{Q(x)} y = 0.$

$$-\int P(x)\,dx = \int \frac{+2x}{1-x^2}\,dx = -\ln(|1-x^2|)$$

$$e^{-\int P(x)\,dx} = \frac{1}{1-x^2} \qquad x \in (-1,1)$$

$$v(x) = \int \frac{1}{y_1^2} e^{-\int P(x)\,dx}\,dx = \int \frac{1}{x^2} \cdot \frac{1}{1-x^2} \cdot dx$$

$$v(x) = \int \frac{1}{x^2} + \frac{1}{1-x^2}\,dx = -\frac{1}{x} + \frac{1}{2}\ln\left(\left|\frac{1+x}{1-x}\right|\right)$$

$$y_2(x) = v(x)\,y_1(x) = -1 + \frac{x}{2}\ln\left(\left|\frac{1+x}{1-x}\right|\right) ; \; x \in (-1,1).$$

Solution of Second-order Homogeneous ODE with constant coefficients

given $ay'' + by' + cy = 0$. (or) $y'' + py' + Qy = 0$.

$$y'' + 2\xi\omega_n y' + \omega_n^2 y = 0.$$

$\xi \leftarrow$ damping factor

$\omega_n \leftarrow$ natural freq.

Guessing the sol: e^{sx}.

$\hookrightarrow$ True if $\underbrace{(s^2 + 2\xi\omega_n s + \omega_n^2)}_{=0}\, e^{sx} = 0$.

2 roots $= -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$

roots are real & distinct if $\xi^2 - 1 > 0 \Rightarrow s_1$ and s_2.

$y_1 = e^{s_1 x}$ $\quad y_2 = e^{s_2 x}$ $\quad W(y_1, y_2) = e^{s_1 x} s_2 e^{s_2 x} - s_1 e^{s_1 x} e^{s_2 x}$

$= (s_2 - s_1)\, e^{(s_1 + s_2)x}$

gen sol: $y(x) = c_1 e^{s_1 x} + c_2 e^{s_2 x}$.

roots are Complex: if $|\xi| < 1$: $s_1 = \underbrace{-\xi\omega_n}_{\alpha} + j\underbrace{\omega_n\sqrt{1-\xi^2}}_{\omega_d}$ $\quad s_2 =$

$\omega_d \rightarrow$ damped freq of oscillations.

$\alpha \hookrightarrow$ rate at which the transient part will decay

$y(x) = c_1 e^{s_1 x} + c_2 e^{s_2 x}$ where $s_2 = s_1^*$

$\underline{\underline{c_2 = c_1^*}}$ (sol needs to be real)

$= \underbrace{r\, e^{\alpha x} \cos(\omega_d x + \phi)}$ $\leftarrow$ $= c$. is a Complex Const.

Solution of Second-order Homogeneous ODE with constant coefficients

roots repeat if $|\xi| = \pm 1$ (Critical damping):

$$s_1 = s_2 = \xi\omega_n = \pm\omega_n.$$

$$y(x) = c_1 e^{s_1 x} + c_2 x\, e^{s_1 x}$$

in normal form: $y'' + 2\omega_n y' + \omega_n^2 y = 0.$

roots are $-\omega_n$ (mul $= 2$)

$y_1 = e^{\omega_n x}$ is a sol. $\therefore$ $y_2 = v(x)\, y_1$

$$v(x) = \int \frac{1}{e^{-2\omega_n x}} \cdot \underbrace{e^{-\int 2\omega_n dx}}\, dx = \int dx = \underline{\underline{x}}.$$

$$y_2 = \underline{\underline{x}}\, e^{-\omega_n x}.$$